2. Formatting

Formatting functions

`#<func>(<key: value list>)[content]`:

- The parentheses are for settings; the list is in code mode → no # is allowed.
- The brackets are for whatever is affected by the settings, they are in markup mode → # is required for a function.

Here are some functions of such a form with some of their settings:

- **`text`** – sets font family, size, color, etc. of the text.
  - `font: "<fontname>"`
  - `size: <number>pt`
- **`page`** – sets page size, margins, headers, enable columns and footers.
  - `paper: "a5"`
  - `margin: (<key: value list>)`
    - `x: <length>`
    - `y: <length>`
    - `left: <length>`
    - `right: <length>`
    - `top: <length>`
    - `bottom: <length>`
- **`par`** – sets paragraph justification, line spacing, etc.
  - `justify: true`
  - `leading: <length>` – space between lines
- **`heading`** – sets the appearance of headings and enable numbering.

  - `numbering: <string>` – can use letters, roman numerals and symbols.
    - "1." ⟶ 1. … 1.1. … 1.2. … 2. … 3.2. …
    - "1.a" ⟶ 1. … 1.a … 1.b … 2. … 3.b …
- **`document`** – sets the PDF metadata, such as title and author.
- **`align`**
  - `<horizontal alignment>`, e.g., center
  - `<vertical alignment>`, e.g., `bottom`
  - The alignments can be combined with + sign.

To affect formatting of all the following document, use **`#set <function>(...)`**.

`#show <element>: <var> => ...#<var>...` change all occurrences of the `<element>` (now represented by `<var>`) in the document into the `...#<var>...` To affect a text, use `#show "<text>"...`

Offtop tips

- `#lorem(<number>)` produces `<number>` words of dummy text.
- To turn autocomplete panel off, use `Esc`. To turn it on, use `Ctrl + Space`.
- `em` is a relative length equal to the current font size.
- `#image(height: <length>, ...)` produces an image of specified height.
- Use `#box` to prevent undesired line breaks, e.g., `#box[#box(image(...)) <text>]`.

Questions

1. Suppose a function has this form: `#<func>(...)[...]`. What goes into (...) and [...] respectively? Where we use `#` and where we don't, and why so?
2. Explain each functions:
   - `text`
   - `page`
   - `par`
   - `heading`
   - `document`
   - `align`
3. Explain each argument. Which function it belongs to, what does it control, which values does it take?
   - `font`
   - `size`
   - `paper`
   - `margin`
   - `justify`
   - `leading`
   - `numbering`
   - center
   - bottom
4. When we combine two arguments with +?
5. Explain the `#set` construction.
6. Explain the `#show` construction.
7. Explain the `#lorem` function.
8. Explain the `em` unit.
9. How to make an image have a certain height?

10. Explain the following construction:
    `#box[#box(image(...)) <text>]`.
11. How to turn autocomplete panel on and off?
12. Explain in terms of functions, function arguments and argument values:
    - How to switch font to New Computer Modern?
    - How to switch font to 14pt?
    - How to change page size to A5?
    - How to make horizontal margins 1.5 centimeter wide and vertical 1 centimeter wide?
    - How to make text fully justified?
    - How to make space between lines 1.2 of font size?
    - How to make headings be numbered like that: 1. … 1.1. … 1.2. … 2. … 3.2. …
        - How to make heading be numbered like that: 1. … 1.a … 1.b … 2. … 3.b …
        - How to make an element (text/picture/figure, etc.) aligned to the center? to the bottom? to the center and the bottom simultaneously?
        - How to change PDF metadata.
13. How to change all occurrences of ABC in the document into DEF?
14. How to add 15 words of Lorem ipsum into the document?
15. How to prevent undesired line breaks, e.g., before and after an image?